

HW06 API Testing Main Report

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****Generated at:**** 2026-08-22 22:30 Asia/Saigon

****Latest local execution:**** 2026-08-24 00:38-00:39 Asia/Saigon

Scope

Three selected APIs were tested from three different feature pools:

Pool	Feature	Endpoint(s)	Test cases
A	FR-03 Password recovery	POST /api/forgot-password, POST /api/reset-password	42
B	FR-09 Apply coupon	POST /api/apply-coupon	37
C	FR-13 Admin orders	GET /api/admin/orders	36

Pipeline

1. Read api_specification.md and backend/server.js.
2. Generate AI draft cases by equivalence partitioning, boundary value analysis, security testing, schema validation, state transition testing, fuzzing, and error guessing.
3. Human-audit each AI case as VALID / INVALID / INCOMPLETE and correct the weak cases.
4. Add five human cases per API, especially for security and workflow risks.
5. Export Excel workbooks and Postman/Newman execution artifacts.

Test Case Summary

API	AI VALID	AI INVALID/INCOMPLETE	HUMAN VALID	Total
FR03	33	4	5	42
FR09	29	3	5	37
FR13	27	4	5	36

Postman Features Used

Feature	Used	Evidence
Collection	Yes	postman/hw06-api.postman_collection.json
Environment variables	Yes	baseUrl, studentId

Feature	Used	Evidence
Collection folders	Yes	One folder per selected API
Pre-request scripts	Yes	Adds X-Student-Id, prepares dynamic bodies, logs in users
Test scripts	Yes	Status, content type, response time, schema-level checks
Data-driven runs	Yes	fr03-data.json, fr09-data.json, fr13-data.json
Newman CLI	Yes	Commands documented in run-notes.md
Mock servers	No	Real local SUT was used
Monitors	No	Local and CI execution are the required targets

Bugs Found

Bug ID	Endpoint	Summary	Severity
BUG-FR09-001	POST /api/apply-coupon	Percent coupon formula computes negative discount.	High
BUG-FR13-001	GET /api/admin/orders	Any valid user token can access admin order list.	Critical
BUG-FR09-002	POST /api/apply-coupon	Public endpoint trusts body user_id for usage-sensitive logic.	Medium

Detailed bug report is in bug-reports/bug_report.md. GitHub issue links/screenshots are marked TODO because real issue evidence must be created manually and cannot be fabricated.

Execution Result

Local Newman execution was completed on 2026-08-24 00:38-00:39 Asia/Saigon against http://localhost:3000 with X-Student-Id: 23127296 injected by the pre-request script.

API	Iterations	Requests	Assertions	Failed Assertions
FR03	42	68	168	0
FR09	37	37	148	0
FR13	36	68	144	0
Total	**115**	**173**	**460**	**0**

JSON reports are stored in newman/fr03-newman-report.json, newman/fr09-newman-report.json, and newman/fr13-newman-report.json.

Remaining Evidence To Complete

The local API execution evidence is complete. The following items still require real external evidence and must not be fabricated:

- GitHub Issues for the documented bugs, each with a screenshot and issue URL.
- GitHub Actions passing run link and screenshot after pushing the HW06 workflow.
- GitHub Actions failing run link and screenshot from a temporary secure-oracle failing commit.
- Optional Newman HTML reports if the grader expects HTML in addition to the generated JSON reports.
- A manually confirmed self-drawn/exported generator diagram image and optional demonstration video link.

AI Audit Declaration

I use AI tools for the following tasks: generating initial test-case ideas, auditing against source code, designing Postman data-driven automation, drafting the generator skill, and preparing Markdown report templates. Full interaction log is summarized in ai_audit_report.md.

AI Critique

See ai_critique.md.